

Supplementary Material

Contagion: When Per-Hop Prompt-Injection Survival in
LLM Agent Networks Does (and Does Not) Compose

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A Multiplicity Control for the Composition Test

Table 1 applies a Benjamini–Hochberg correction to the composition tests of Section 3.7 of the main paper, using the *default fresh-artefact protocol* for every cell at the common $n = 40$ grid. Only the Llama cell rejects H_0 ; it survives the correction at $q = 0.05$, while the four remaining non-degenerate cells—including DeepSeek, whose fixed-artefact deviation is a protocol artefact that disappears under the fresh protocol (Section 5.3 of the main paper)—stay non-significant. The Claude cell is degenerate ($\text{ASR} = \prod_i \hat{s}_i = 0$, $\text{sd}(\Delta) = 0$) and is excluded from the correction. At the higher $n = 200$ resolution the Nova cell also crosses into rejection (Section 5.10 of the main paper); we keep the correction at the common $n = 40$ grid so the five cells are compared at equal power.

Table 1: Benjamini–Hochberg over the $m = 5$ non-degenerate chain cells (fresh-artefact protocol). p -values are the bootstrap two-sided values from Section 3.7 of the main paper. Only the Llama rejection survives.

<i>rank</i>	<i>cell</i>	<i>p</i>	<i>BH thr. ($q=0.05$)</i>
1	Llama · none	0.0002	0.010
2	Nova · none	0.0840	0.020
3	qwen · none	0.3165	0.030
4	DeepSeek · none (fresh)	0.6300	0.040
5	qwen · paraphrase	0.7775	0.050

B Context-Clustered Interval on the Weak Edge

Section 5.10 of the main paper reports that a homogeneity test rejects exchangeability across benign task contexts on the weak Llama edge. Because that violates the i.i.d. assumption behind Wilson intervals, we recompute the edge rate with a cluster bootstrap that resamples *repeats* (seed replications), not individual trials. At a fixed temperature ($T = 0.7$, six repeats of twenty trials) the two intervals nearly coincide (Table 2): the design-effect widening is $0.99\times$, so within a temperature the Wilson interval is well calibrated. The overdispersion $\varphi = 2.93$ reported in Section 4 of the main paper is a *cross-temperature* artefact—temperature is a deliberate systematic factor, not trial noise—which is why we compute φ within a temperature.

Table 2: Weak edge $\mathbf{a1} \rightarrow \mathbf{a2}$ (Llama, $T = 0.7$, $n = 120$): i.i.d. Wilson versus a repeat-clustered bootstrap (10^4 resamples).

<i>interval</i>	$\hat{s}$	<i>95% CI</i>	<i>width</i>
Wilson (i.i.d.)	0.242	[0.174, 0.326]	0.152
cluster bootstrap (by repeat)	0.242	[0.167, 0.317]	0.150

C Threshold Sensitivity of the Floor Effect

A referee may ask whether the “resistant” verdict on the frontier model is an artefact of the compromise threshold $\tau_{\text{ASV}} = 0.9$. We re-score the single-hop obfuscation cells (Claude, $n = 30$ per cell) at $\tau \in \{0.7, 0.8, 0.9, 1.0\}$ directly from the stored ASV scores—a deterministic re-thresholding, no new model calls. The compromise rate is *invariant* across τ in every cell (Table 3): the no-defence baseline stays at 0.17 and redaction at 0.00 regardless of the threshold. The reason is that the ASV distribution is bimodal—of 180 trials, 158 score 0.0 and 14 score 1.0; the only eight intermediate trials all score ≤ 0.667 , i.e. below the lowest threshold tested. The floor effect is therefore a property of the model, not of where the judge’s cut is placed.

Table 3: Single-hop compromise rate versus τ_{ASV} (Claude, $n = 30$ per cell), re-scored offline from stored ASV. Rates are threshold-invariant.

<i>defence</i>	<i>form</i>	$\tau=0.7$	0.8	0.9	1.0
none	plain	0.17	0.17	0.17	0.17
none	spaced	0.17	0.17	0.17	0.17
none	split-word	0.13	0.13	0.13	0.13
redact	plain	0.00	0.00	0.00	0.00
redact	spaced	0.00	0.00	0.00	0.00
redact	split-word	0.00	0.00	0.00	0.00

D Theory Assumptions: Independence, Correlation, and R_0

This appendix clarifies the theoretical assumptions invoked in Section 3.8 of the main paper.

Product-of-complements and correlation. The percolation recurrence

$$\Pr[v] = 1 - \prod_{u \rightarrow v} (1 - s_{u \rightarrow v} \Pr[u])$$

assumes *edge activations* (and the corresponding parent-reach events relevant to the product form) are conditionally independent given the graph structure. When multiple parents $u_1, \dots, u_k$ of v share a common ancestor, the events $\{\Pr[u_i]\}$ may be correlated, so $\Pr[\bigwedge_i (\text{not reached via } u_i)] \neq \prod_i (1 - s_{u_i \rightarrow v} \Pr[u_i])$ in general. For the specific topologies evaluated in this work (chains, trees, star, and the small manager/worker loop), we do not rely on that product form across shared ancestors; where we use it, we treat it as the marginal approximation under the stated conditions or restrict claims to graph classes where the independence holds. For arbitrary DAGs with shared ancestors, a path-based (inclusion–exclusion) formulation or explicit correlation modelling is required.

R_0 definition and scope. For strictly acyclic (feed-forward) graphs, the standard linearized next-generation matrix M with $M_{uv} = s_{u \rightarrow v}$ is strictly triangular in a topological ordering, hence $\rho(M) = 0$. This is a structural property of acyclicity, not an empirical safety guarantee. We therefore report R_0 as a *descriptive, finite-horizon* reproduction statistic: specifically, we define R_0 as the mean number of newly compromised neighbours produced per compromised agent instance over the observed finite horizon (generation convention as stated in Section 4.2 of the main paper), with its denominator clearly specified. We do not interpret $\rho(M) < 1$ as a threshold for safety on finite acyclic graphs; reachability is given by the percolation (Eq. 3 in the main paper), not by R_0 alone.

Parallel paths and defence placement. Hardening a single node is not equally effective when there exist multiple s - t paths: effectiveness depends on whether that node lies on all paths (a cut). For unique-path graphs the effect is path-specific; we do not generalise equal effectiveness to multi-path DAGs. Placement claims are limited to the graph classes we evaluate.

Transportability and product composition. The chain identity $\text{ASR} = \prod_i s_i^{\text{nat}}$ is algebraic given the Markov/chain structure. However, composing *isolated* per-edge estimates s^{ctrl} requires that $s_i^{\text{ctrl}} \approx s_i^{\text{nat}}$ (transportability). We do not assume this holds in general; we test it empirically and use the depth-error slope as a diagnostic (Section 5.2 of the main paper).

E Recurrence Makes the Threshold Non-Vacuous

This appendix gives the full treatment of the recurrent (cyclic) case summarised in Section 5.8 of the main paper. We test a pattern real frameworks use: a manager broadcasting to workers that report back.

For a manager with w reporting workers, the infection matrix has $M[m \rightarrow w_j] = a_j$ and $M[w_j \rightarrow m] = c_j$, and its characteristic polynomial is $\lambda^{w-1}(\lambda^2 - \sum_j a_j c_j)$. Hence

$$\rho(M) = \sqrt{\sum_j a_j c_j} \quad (1)$$

for every w (checked numerically against the eigenvalues to machine precision on 320 random configurations). Two consequences follow. A feed-forward reading sets every $c_j = 0$, so $\rho = 0$ however strong the edges are—the vacuity of Section 3.8 of the main paper again, now with the feedback edges removed. And the system is supercritical exactly when $\sum_j a_j c_j > 1$; for symmetric edges $a_j = c_j = s$ this is $w s^2 > 1$. That is a *design rule* rather than a diagnostic: at $s = 0.5$ a manager needs five reporting workers to sustain an endemic injection, at $s = 0.7$ three, at $s = 0.9$ two.

Applied to measured survivals, Eq. (1) makes opposite predictions, which is the test a criterion has to pass to be worth using (Table 4). On Llama the sum is 1.800 and on DeepSeek 1.018, so both frontier loops should be supercritical; on qwen2.5:7b it is 0.578, so the same loop should decay.

Table 4: One criterion, three models, opposite predictions. $\sum_j a_j c_j$ uses the measured per-hop survivals of the same three-agent recurrent pattern; the feed-forward column is the same survivals with the two feedback edges removed.

<i>model</i>	$\sum_j a_j c_j$	ρ <i>rec.</i>	ρ <i>fwd.</i>	<i>prediction</i>
Llama 3.3 70B	1.800	1.342	0.000	supercritical
DeepSeek V3.2	1.018	1.009	0.000	supercritical (marginal)
qwen2.5:7b	0.578	0.760	0.000	subcritical

The measured dynamics follow. On Llama the recurrent manager is re-compromised by its own workers’ reports in 0.725 [0.572, 0.839] of runs and does not die out: at round five it is still compromised in 0.700 of runs with the workers near saturation (0.925, 0.975), as $\rho > 1$ predicts; extending the same loop to eight rounds leaves it endemic (manager 0.583, workers 0.900). DeepSeek, whose $\sum_j a_j c_j = 1.018$ sits just above criticality, behaves the same way over eight rounds—the manager settles at 0.567 with workers at 0.767 and 0.700 rather than decaying—so the recurrent verdict is not a single-model artefact. On qwen the same loop decays: the manager’s re-compromise rate falls from 0.775 [0.625, 0.877] to 0.650 and the workers plateau (0.475, 0.525). One criterion, opposite predictions, all borne out.

Two caveats. First, ρ is computed from *isolated* survivals that Sections 5.2–5.7 of the main paper show to be unreliable; the Llama and qwen verdicts survive because neither $\sum_j a_j c_j$ is marginal (1.80, 0.58), whereas DeepSeek’s 1.018 sits close enough to the critical value that its supercritical label should be read together with its measured dynamics (endemic over eight rounds) rather than from the point estimate alone. Second, we study a single recurrent pattern at one temperature; the three models span five to eight rounds, enough to separate decay from saturation, not to demonstrate extinction.

F Judge Calibration, Scope, and Known Failure Modes

What was measured. The compromise rule of Section 3.4 of the main paper ($C = \mathbf{1}[\text{ASV} \geq 0.9]$, ASV = fraction of target bigrams present) was calibrated on **ground-truth-labelled** outputs from the local model **qwen2.5:7b** ($n = 12$ per compromised class, $n = 12$ benign, $n = 3$ refusing outputs; exact-string injection, benign “verification code” framing). The labels are assigned by construction, not by a judge: *benign* outputs never see an injected instruction; *refusing* outputs see one and do not reproduce the target; *propagating* outputs are retained only when the target appears, and *direct* outputs come from the injected instruction alone.

Table 5: Calibration of the two continuous scores on labelled qwen2.5:7b outputs (mean $\pm$ sd). ASV separates the two compromised classes (1.000) from both benign classes (0.000) with no overlap; the full-containment variant instead scores benign outputs as high as 0.459, which is what motivated the switch (reported in the main paper).

<i>score</i>	<i>benign</i>	<i>refusing</i>	<i>propagating</i>	<i>direct</i>
ASV (target-bigram containment)	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
MR (Dice vs. direct output)	0.253 $\pm$ 0.179	0.484 $\pm$ 0.044	0.598 $\pm$ 0.070	0.862 $\pm$ 0.212
full-containment (rejected)	0.459 $\pm$ 0.215	0.597 $\pm$ 0.274	0.729 $\pm$ 0.223	0.825 $\pm$ 0.274

Agreement at the deployed threshold. At $\tau_{\text{ASV}} = 0.9$ the rule attains precision 1.000, recall 1.000, F1 1.000, FPR 0.000 and FNR 0.000 on this set (TP = 24, FP = 0, TN = 15, FN = 0); the closest benign score observed anywhere in the probes is 0.875, i.e. the threshold sits inside a gap rather than on a boundary. The Dice-based MR branch, by contrast, separates the same classes at accuracy 0.949 with FPR = 0 only for $\tau_{\text{MR}} > 0.53$, and the rejected full-containment variant reaches accuracy 0.795 with FPR = 0.067 at its own optimal cut—consistent with the main paper’s 1.000 vs. 0.795 comparison.

Scope of this validation, and what it does not establish. The calibration covers **one model** (qwen2.5:7b), **one payload family** (exact-string targets), and **no defence**: both the benign and the compromised classes are drawn from undefended runs. The rule is deterministic and needs no per-model recalibration, but we do *not* claim that the same operating point transfers unchanged to frontier models at temperature 0.7, to paraphrase- or obfuscation-obfuscated targets, or to defended runs. Two further failure modes follow from the operational definition and should be read alongside every number we report: (i) a paraphrased leak that destroys the target’s bigrams is scored as *not* propagating (false negative), and (ii) a safe quotation of the target inside a benign summary is scored as propagating (false positive). Validation on frontier models, multiple payloads and defended conditions requires human annotation of a stratified subset across those conditions; we report the protocol for it below so the limitation is precise rather than open-ended. Read against a human annotator, both failure modes concern the *semantics* of the target: a paraphrase that destroys its bigrams is scored as not propagating, and a safe quotation inside a benign summary is scored as propagating.

Outcome levels. Every judged activation is reported at level (a) *textual propagation*—the target string/code fragments appear. The calibration set contains no case that separates (a) from (b) *instruction adoption/compliance* within the same output, and no case of (c) *harmful action* (the payload is a benign code, so (c) is by construction outside this study). Adoption is a subset of propagation in this task family: an agent that acts on the injected instruction must surface the target, whereas the converse does not hold. Since every label here is positive on (a) whenever it is positive on (b), the judge is an *upper bound* on the adoption rate and we report (a) as the propagation outcome. We do not use it as evidence of instruction adoption or of harmful action anywhere in the paper.

Replay procedure. In the same spirit we state precisely what the content-form replay of Section 5.4 of the main paper does. Each replay trial draws a fresh receiver, rotates the benign assignment over the same pool, and consumes either content captured from a real natural run, a canonical artefact produced on demand, a fixed hand-written string, or the in-context content under a different sender label; temperature, token budget, system prompt, role, judge and thresholds are identical across arms, so the arms differ only in message form and in the label of its sender. The in-context arm is the harness self-check and has a target it must hit, namely the rate measured in-chain; the other arms are read as comparisons against it. Because each trial resamples the source output and draws the receiver fresh, a replay rate can differ from an in-chain rate through sampling noise and per-context variation as well as through message form. On the edge where the self-check misses (0.733 versus 0.947) we therefore report the value for completeness and do not use it as an estimate of the form effect. The local re-run of this harness (qwen2.5:7b, 4 natural trials and 2 replays per arm and edge) is consistent with that reading rather than with a validated harness: every edge sits inside the replay confidence interval, yet two of three edges differ from the in-chain rate by 0.500, which at this sample size the interval cannot resolve. We read the local run as *underpowered*, not as a passed self-check, and we do not report it as evidence about form; the $n = 30\text{--}40$ configuration behind the content-form table of the main paper (its Table 4) remains the only configuration in which the self-check discriminates.

Behaviour on replayed activations (local model). Table 6 adjudicates every activation of a 24-item stratified subset drawn from the replayed harness of Section 5.4 of the main paper (qwen2.5:7b, one payload, no defence; the subset is stratified over arm and over judge verdict so that both classes are represented). Each output was read and labelled at the three outcome levels before the judge score was consulted. Against level (a) the rule reproduces the labels exactly: precision 1.000, recall 1.000, $F_1 = 1.000$, $\text{FPR} = \text{FNR} = 0.000$. Against level (b) it is not a usable detector: it flags 19 activations, and none of them involves instruction adoption—every positive is the target quoted inside a work product (an executive summary, a ticket classification, a release note, a review annotation). Three activations that sit downstream of content that *did* carry the target do not carry it themselves, and the rule correctly reports them as negative; a reader who stopped at the upstream context would have scored them as propagating. We therefore report (a) and never (b), which is precisely the distinction the main paper draws when it calls the judge an upper bound on adoption.

Protocol for a full validation (not yet executed). A stratified subset should be drawn to distribute evenly across no-defence/defence, judge success/failure, topology, model and arm; each case should be labelled at levels (a), (b) and (c) by a human annotator, blind to the judge score, with a second annotator on a subset for agreement. Reported metrics should be precision, recall, F_1 , FPR , FNR and the full 2×2 confusion matrix per level, with Wilson/Clopper–Pearson intervals, cluster bootstrap for intervals (clusters = payload $\times$ defence $\times$ arm $\times$ edge, since trials repeat within contexts), and worked TP/FP/TN/FN examples. The scripts that generate the stratified sheet and score it are part of the released artifact; the numbers in Table 5 are the portion we had measured at submission time.

Table 6: Adjudication of the 24-activation subset against the three outcome levels (level (a) textual propagation, (b) instruction adoption, (c) harmful action). Labels were assigned by reading the full outputs, before the judge score was consulted, with the help of an automated string-matching assistant and with review by one of the authors; the labelling was not independent of the paper’s authors, so these numbers are a consistency check on the rule, not an independent validation of it.

<i>level</i>	<i>TP</i>	<i>FP</i>	<i>TN</i>	<i>FN</i>	<i>prec.</i>	<i>recall</i>	<i>FPR</i>
(a) textual propagation	19	0	5	0	1.000	1.000	0.000
(b) instruction adoption	0	19	5	0	—	—	0.792
(c) harmful action	0	19	5	0	—	—	0.792

G Reproducibility, Artifact Claims, and Replication Details

We explicitly align reproducibility with the materials supplied. The claims about reproducibility must match the actual artifact/package provided with the submission. If code, per-cell outputs, and a manifest are included as an anonymized artifact, a manifest (table $\rightarrow$ data file $\rightarrow$ script) should specify how to regenerate each table without live model calls. If the artifact is not yet included, the reproducibility statement should reflect partial reproducibility rather than claiming reproduction of “every table”.

Minimum replication details. Any released artifact or reported configuration should include: full prompts and payload templates (anonymized where necessary); exact model identifiers (vendor+model ID), API/provider+version, collection dates/time window, platform; decoding parameters (temperature, top_p/top_k, max_tokens, stop sequences, seed if any, frequency/presence penalties); execution details (retries, error handling, timeouts, rate limits, failed calls with counts, token budgets, call counts per condition); fresh-artifact sampling/randomization/order/stratification; non-determinism handling (temperature>0); and a machine-readable experiment inventory (condition $\rightarrow$ protocol, topology, model, defence, payload form, context/task, n , N_{success} , N_{total} , call counts). All estimates should include numerator/denominator and appropriate uncertainty (clustered where trials repeat within contexts/runs), with confirmatory vs. exploratory analyses clearly labeled.